Matej Ciglenečki

CTO with over 4 years of experience, building a distributed machine learning system serving tens of thousands of active users. My research interests include **video understanding, image editing, and 3D/4D reconstruction**, with openness to related areas**. Actively preparing to pursue a machine learning PhD, with a primary goal of publishing in top-tier ML conferences.** Throughout my experience, I've developed strong research and problem-solving skills, with a clear sense of ownership and the ability to communicate technical findings effectively.

Zagreb, Croatia
matej.ciglenecki@gmail.com
matejciglenecki.com
github.com/ciglenecki
linkedin.com/in/matej-ciglenečki

Education

M.Sc Data Science – Faculty of Electrical Engineering and Computing, Zagreb, Croatia — Oct 2021 – July 2023

relevant courses: Machine Learning, Deep Learning, Deep Generative Modeling, Multivariate Data Analysis, Signal processing, Natural Language Processing, Numerical Mathematics, Advanced Algorithms and Data Structures, Parallel Programming

master's thesis: **Generative Novel View Synthesis Using Neural Radiance Fields (NeRF).**

- During my collaboration with Microblink, I built a model for generating personal document images, identified constraints for high-quality image generation, and quantified image quality in angle interpolation and extrapolation setups. I also developed a segmentation mask generation method via transfer learning to minimize the required annotation data, and created an automated cropping method to isolate documents in space and reduce image artifacts.

B.Sc Computer Science – Faculty of Electrical Engineering and Computing, Zagreb, Croatia — Oct 2017 – July 2021

relevant courses: Algorithms and Data Structures, OOP, Design Patterns, Databases, Probability and Statistics, Artificial intelligence

bachelor's thesis: **Facial Expression Recognition Using Convolutional Neural Networks**

- Trained an EfficientNet to classify human emotions from image sequences using scraped and labeled data. Scaled sequence labels so the model could more accurately learn the change from neutral expression to the final emotion.

Experience

TensorPix – CTO — Dec 2024 – present

- Trained a super resolution model conditioned on the degradation embedding that's currently in production (**PyTorch**).
- Built and trained a model that predicts an image's degradation embedding (blur, compression, noise…) (**PyTorch**).
- Built a high-quality image filtering pipeline to filter training data, improving the super resolution model quality metrics.
- Developed a custom video inference engine for efficient video processing (**PyTorch, Python, ONNX, TensorRT**).
- Researched and benchmarked and torch.compile for the purpose of inference cold start with static and dynamic input shapes.
- Enhanced generative modeling benchmarking and evaluation suite (**PyTorch, Python**).
- Managed 3 engineers. Heavily influencing the company's strategy. Pitched to investors during Series A fundraising.

TensorPix – Machine Learning Engineer — Nov 2023 – Dec 2024

- Researched, implemented, and trained models for super resolution and low-light video enhancement (**PyTorch, Python**).
- Implemented a task queue and distributed workers for ML inference used in production (**Redis, Python, Django**).
- Implemented a custom cloud-agnostic instance auto scaler (**Redis, Python**).

Microblink – Machine Learning Engineer Intern — Feb 2023 – Sep 2023

- Researched and implemented NeRF to support the generation of segmentation masks via transfer learning. (**PyTorch**)
- Developed NeRF scene cropping method to isolate the central object to reduce foreground image artifacts. (**PyTorch**)
- Processed point clouds and images, numerically validated view extrapolation, mesh reconstruction, generalization across different scenes and correlation between scene's variability and reconstruction quality (**Python, PyTorch**).
- Researched and implemented a font classifier via semi-supervised learning used for document verification (**PyTorch**).

Photomath (Google) – Software Engineer Intern — July 2022 – Oct 2022

- Developed cloud services that parse, transform, enrich and deliver millions of events from Photomath's large textbook solutions database (Python, GCP, Datastream, Pub/Sub, Dataflow, Cloud Run, GitHub Actions, Spark).

Memgraph – Software Engineer — July 2020 – Oct 2021

- Designed and implemented the database schema and backend for Memgraph Cloud. It manages AWS EC2 instances, and supports monthly user billing based on user usage (node.js, TypeScript, Express, Sequelize, PostgreSQL).
- Set up observability Elastic Stack on AWS EC2 to analyze application logs via AWS CloudWatch in Kibana dashboards.

Publications

A Squeeze of LIME, a Pinch of SHAP: Adding the Flavor of Explainability to Sentence-Level Commonsense Validation
(M. Moslavac, R. Grbelja, M. Ciglenečki), Text Analysis and Retrieval 2023 Course Project Reports

- Trained a DeBERTa-based model for classification of sentence coherence.
- Analyzed explainability of the trained model with LIME and SHAP, model-agnostic and post-hoc methods.

Extracurricular

Torch JPEG Blockiness Metric 2025

- Built a PyTorch JPEG blockiness metric to detect low-quality JPEG compression in production image datasets at TensorPix.
- Vectorized the algorithm to support fast batched image scoring on same-size inputs.
- Validated the implementation against the original paper implementation.

Machine Learning competition – Musical Instrument Classification 2023

- **Led a team of four. Won the 2nd place and 1st place in model performance.**
- Trained a model that labels 11 musical instruments from an input audio signal, developed in PyTorch using deep learning, digital signal processing, audio feature engineering, and computer vision style spectrogram/image representations derived from raw audio. Deployed the trained model behind a FastAPI inference service.
- Wrote both competition report and technical documentation.
- LUMEN Data Science is the largest machine learning competition in the Croatia.

StyleGAN2 latent projection inversion 2022

- Built latent space tools for W+ projection, reconstruction and interpolation videos.
- Trained and evaluated StyleGAN2 and StyleGAN2-ADA across full-data and low-data regimes.
- Built a PyTorch pipeline for FFHQ-style preprocessing, face alignment, and normalization.

Machine Learning competition – Computer Vision GeoGuessr 2022

- **Led a finalist team of three. Won the 2nd place in model performance.**
- Trained a model that predicts the lat/lng location of Google Street View images, developed in PyTorch using deep learning, computer vision, and geospatial feature engineering. Deployed the trained model behind a FastAPI inference service.
- Wrote both competition report and technical documentation.
- LUMEN Data Science is the largest machine learning competition in the Croatia.

Paper implementation – driver fatigue detection in an EEG-based system 2022

- Reproduced paper's baseline and methods. Trained the model that matches the reported performance.
- Analyzed and extracted entropy features from driver's EEG data.
- Trained and compared four different models, achieving +1% higher performance than reported in the paper.

Skills

Languages:	**Python, TypeScript, bash, C++**
Technologies:	**git, Linux, PyTorch, OpenCV, numpy, Docker, LaTeX**
Other:	**Machine Learning, Computer Vision, Generative Models, Distributed Systems, NLP**

Other

Certificate in Advanced Rhetoric and Public Speaking (Heureka) 2026
Certificate in Rhetoric and Public Speaking (Heureka) 2025
Attended Croatian Machine Learning Workshop (CMLW) 2024
Certificate in Soft Skills (EESTEC) 2019